LESSON PLAN

EMI/EMC (14)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	05
Total No. of Lesson Plans	46	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Introduction to the navigation components of the triangle of velocities i.e. airspeed, Mach nos., ground speed, track and drift.
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Introduction to the navigation components of the triangle of velocities i.e. airspeed, Mach nos., ground speed, track and drift. Introduction to the application of the triangle of velocities in navigation.
5. Specific Objectives	14.1.1.1 Trainee will be able to explain and define a position on the earth.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of vector addition concepts using PPT diagrams; Safety: Ensure no loose items on desk; Supervision: Instructor monitors diagram interpretation.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the definition of a position on the earth?
(b) Define the term 'airspeed' in the context of navigation.
(c) How does wind affect an aircraft's ground track?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Definition of Airspeed • True Airspeed vs Indicated Airspeed • Mach Number and Speed of Sound • Ground Speed Calculation • Track and Drift Angles	L & Ds	PPT, IB	15 min
(b)	• Triangle of Velocities Construction • Vector Addition Principles • Calculating Track and Drift • Effect of Wind on Navigation • Application in Flight Planning	L & Ds	PPT, PDF	20 min
(c)	• Summary of Navigation Components • Relationship between TAS, GS, and Wind • Importance of Accurate Velocity Data • Review of Key Definitions • Preparation for Position Lines	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain how ground speed differs from airspeed.
(b) Describe the relationship between track, heading, and drift angle.
(c) Why is Mach number critical at high altitudes?

12. Home Assignment and Follow Up	Prepare a brief note on how position lines derived from ground D/F stations and TACAN will be used in conjunction with velocity triangle data in the next lesson.
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Introduction to position lines, relative bearing, transit bearings, line features, and position lines from ground D/F stations and TACAN and VOR bearings.	Refer to the master syllabus reference list	LP 06

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min